Feedback Is All You Need

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Abstract

Can a small controller acquire high capability by repeatedly incorporating feedback? We test restricted forms of this hypothesis across three diagnostic studies. In changing classification tasks, a 161-parameter learner with accurate feedback outperforms a 2,561-parameter frozen learner by 52.91 percentage points on linear rules and 37.32 on XOR rules. When both learn, the larger model instead leads by 10.93 points on XOR. In 24 hidden-command languages, retaining correct examples yields no measured compositional benefit for the smallest of four pretrained language models under first-token scoring; low-rank updates improve direct predictions without clear compositional transfer. A follow-up on 24 new environments demonstrates interface sensitivity: 7B oracle-offset composition accuracy rises from 24.74% under first-token scoring to 97.27% under full generation. However, 88.28% of its clean-example composition responses reach the 128-token limit, while every 0.5B generated response fails the requested format. The primary small-model interface interaction is -0.52 points (95% CI $[-1.16, 0.11]$). These results support conditional adaptation and expose limits of the tested update and evaluation mechanisms. They establish neither the sufficiency of feedback for general intelligence nor the irrelevance of capacity, priors, or computation.

1 Introduction

The motivating hypothesis is that a compact cognitive core, coupled to an efficient loop of action, outcome observation, correction, and memory, may acquire capabilities disproportionate to its parameter count. Books and other external records suggest an analogy with accumulated human experience. That analogy motivates investigation; it does not establish a mechanism of human intelligence.

The hypothesis must be distinguished from two stronger assertions: that feedback is literally sufficient without representation or inductive bias, and that existing scaling results are wrong. Training already uses feedback through losses, gradients, rewards, and preferences. Scaling-law work measures relationships among loss, capacity, data, and computation [3, 4]; it does not rule out online adaptation. We therefore compare capacity and feedback as interacting resources.

We retain the researcher's chosen title and follow the organizational pattern of *Attention Is All You Need* [2]: background, mechanism, motivation, experimental specification, results, and conclusion. The title is a hypothesis under examination, not an established sufficiency theorem. Chen et al. already used a closely related title in 2023 [5]; we make no priority claim for the phrase or broad feedback-centered framing.

2 Background

The broad motivating idea has a direct precedent in *Reward is enough* [1]. Silver and colleagues hypothesize that abilities associated with intelligence can arise in service of reward maximization in

a rich environment. Their account explicitly includes bounded capacity and real-time computation, and provides no sample-efficiency guarantee. Our experiments examine particular learners and feedback mechanisms; they do not test every agent covered by that conjecture.

Fast adaptation in recurrent state was studied in RL^2 [6] and in-context reinforcement learning in Algorithm Distillation [7]. RL^2 separates a slowly learned update algorithm from fast changes in hidden state. A small runtime controller can therefore embody expensive prior learning. Reflexion combines an evaluator, verbal reflection, and episodic memory [8]; its feedback sources vary across tasks. Self-Refine uses the same model to generate, critique, and revise an answer with task-specific prompts [9], and explicitly requires sufficient instruction-following ability. Our sequential guess condition reproduces neither method.

Huang et al. distinguish intrinsic self-correction from correction supplied with external correctness signals [10]. Their evaluation also stresses inference-cost comparisons and equally informative initial instructions. Test-time compute studies show that some smaller models can outperform larger ones under specific inference allocations [11]. Accordingly, a small-versus-large win alone is not a new account of intelligence. Our current contribution is a reproducible diagnostic comparison of feedback, capacity, and response interfaces, including failed mechanisms and explicit controls. The literature review is targeted rather than exhaustive; exact versions and reading scope are recorded with the repository's source manifests.

3 Agent and Feedback Model

An agent has parameters θ, persistent memory m, and an interaction history h. Its policy selects an action, and an update U incorporates an environmental response f:

$$a_t \sim \pi_{\theta_t}(\cdot \mid o_t, m_t, h_t), \qquad (\theta_{t+1}, m_{t+1}, h_{t+1}) = U(\theta_t, m_t, h_t, o_t, a_t, f_t).$$

An implementation may update parameters, memory, context, or some combination. Its entire system capacity includes external storage and algorithms; a small neural core is not necessarily a small complete system.

Study A instantiates this notation with a binary classifier whose action is a label prediction. Feedback gives the true label after prediction. Inputs are independent of the action, and gradient descent updates the parameters. This is supervised online learning, not active exploration or a new reinforcement learning algorithm. In a binary task, correctness plus the chosen label also determines the target label.

4 What Feedback Can and Cannot Supply

4.1 An information constraint

Let Θ be uniform over $K > 1$ tasks. Assume the complete initial agent state and all side information are independent of Θ. Count every subsequent task-dependent observation as part of the response transcript. If each of n responses has an alphabet of size at most 2^b, then $I(\Theta; F_{1:n} \mid S_0) \leq nb$. Adaptive queries are functions of S_0 and preceding responses, so cannot independently supply missing task information. Fano's inequality gives the exact-task identification bound

$$P_e \geq \max\left\{0, 1 - \frac{nb+1}{\log_2 K}\right\}.$$

Indeed, $H(\Theta \mid F_{1:n}, S_0) \leq 1 + P_e \log_2 K$; subtracting this entropy from $\log_2 K$ yields the result. The bound is about task identification, not every prediction or reward metric. Informative priors or unaccounted observation channels invalidate these assumptions.

For a binary symmetric channel with independent flip probability p, the per-response information is at most $1 - H_2(p)$ bits. At $p = 1/2$, responses carry no information about the true bit. At $p = 1$, inversion is recoverable if the corruption mechanism is known. Learner failure under inverted labels therefore diagnoses a misspecified channel, not destroyed information.

4.2 Representation and transcript constraints

If a successful policy is outside the complete agent's permitted policy class, additional updates within that class cannot express it. External memory or tools may enlarge that class, so this constraint cannot be inferred from neural parameter count alone. In addition, a deterministic updater replaying the same ordered transcript from the same initial state reaches the same final state, by induction. Calling the procedure a loop adds no independent advantage. Active information acquisition or changing observations can matter; these mechanisms require their own tests.

Under the same independent-prior assumption, suppose the final retained state has at most 2^M possible values, with M counting all retained task-dependent bits. Without further task-dependent observations, identification of a uniform task among K possibilities has success probability at most $\min\{1, 2^M/K\}$. For any encoder, the optimal decoder succeeds with probability $K^{-1} \sum_s \max_k P(s \mid \Theta = k) \leq 2^M/K$. This bound concerns exact identification, not every prediction metric. External records belong in the retained state; their capacity cannot be omitted simply because neural parameter count is small. It yields no universal threshold in numbers of neural parameters.

5 Experimental Setup

5.1 Tasks, models, and feedback

Each seed defines an initial and an independently generated post-change rule over eight standard-normal inputs. Linear labels threshold a normalized random projection. XOR labels combine signs of two randomly selected coordinates, with an independent fair complement bit for each rule. Warmup comprises 256 batches of 32 examples; adaptation uses a further 256 batches from the new rule. Evaluation sets of 2,048 examples per rule never enter updates. Separate uniform-input tests preserve each rule and change its input distribution.

The models are logistic regression and one-hidden-layer ReLU networks of widths 16, 64, and 256: 9, 161, 641, and 2,561 parameters. Adam uses learning rate 0.01 and default moments, with fresh moments after warmup. Every arm at the same capacity and seed starts from identical warmed-up parameters and sees the same arriving input stream.

The seven arms are frozen weights, clean labels, 25% flips, 50% flips, inverted labels, clean feedback delayed by 32 batches, and 25% replay. Replay replaces the final eight new examples in every batch with eight warmup examples sampled with replacement, retaining their original labels. It therefore changes both memory use and fresh-example exposure. Delayed feedback has 224 updates at arrival batch 256; a separate flush processes the remaining 32 without new observations.

5.2 Protocol, statistics, and resources

The full grid uses 20 seeds, 4000–4019, and was frozen in repository history before collection. A development run accidentally exposed seeds 1000 and 1001; the seed set was amended before formal collection, without hyperparameter tuning. A subsequent code audit found overlapping PRNG streams in the first full run (seeds 2000–2019). That run is archived with a superseded-inference notice. This version reports a replication on untouched seeds 4000–4019, with streams derived by SHA-256 of environment, family, role, and step or width. A complete-grid collision check passed. The protocol, hyperparameters, and endpoints otherwise remain unchanged. This is repository registration, not external preregistration.

The primary contrast in each family compares width-16 clean to width-256 frozen at arrival batch 256. This deliberately unequal-information comparison tests adaptation, not compute superiority. We report paired seed-level t intervals at 97.5% for each of these two contrasts (Bonferroni familywise 95%). Secondary 95% intervals are descriptive. Seeds, not checkpoints or individual examples, are the replication units; t inference assumes adequate approximation of the seed-mean sampling distribution.

The run used PyTorch 2.8.0 with CUDA 12.6 on one RTX 4090 D and completed 1,120 arms in 109.54 seconds. Code and environment hashes, all 19,200 checkpoint rows, and analysis are retained. For arithmetic-budget comparisons we count $6WN$ multiply-add-related training FLOPs for W matrix weights and N examples. This proxy excludes optimizer, activation, bias, and hardware overhead. Warmup and adaptation are reported separately; it is not lifetime compute or a measured energy budget.

5.3 Study B: pretrained language models

We compare Qwen2.5-Instruct models with 0.494, 1.544, 3.086, and 7.616 billion parameters. Exact Hugging Face revisions and package versions are archived. Each of 24 environments (seeds 3000–3023) assigns four named commands distinct hidden offsets modulo eight. A command adds its offset to an integer state; compositions apply two commands. The family is disclosed, while each environment's offsets are withheld except in an oracle control. Train/test inputs are disjoint: 16 single-command feedback examples, 16 held-out single commands, and 32 held-out length-two compositions per environment. These translations commute, so the test does not measure command-order sensitivity.

Frozen-model conditions are no examples, 4/8/16 correct examples, 16 corrupted examples, 16 sequential model guesses, and explicit oracle offsets. Corruption replaces an answer with a uniformly chosen wrong digit with probability 0.5; this channel still conveys information. Self-generated examples use each model's constrained predictions conditioned only on its earlier guesses. All histories share the same demonstration template. Query selection is fixed, so this is passive adaptation rather than action-dependent exploration. Giving the final correct transcript offline produces byte-identical prompts; we do not count duplicate calls as independent evidence of an online effect. A hand-coded modular-offset estimator solves every test after the same clean examples, demonstrating identifiability within the supplied algorithmic family.

We select the most likely first assistant token among digits 0–7 in bfloat16, without sampling or additional reasoning tokens. Each digit is one tokenizer token. We also record unrestricted first-token accuracy, which does not measure full-response validity. This narrow interface can underuse a model's generative reasoning ability and is a central limitation, not an intelligence metric.

For the 0.5B model, rank-8 LoRA updates all query/value projection matrices (alpha 16, zero dropout), using 16 example-free training prompts for 32 passes, batches of four, AdamW at 2×10^{-4},

and a next-digit cross-entropy loss. Each environment starts from the same base checkpoint and fresh adapters and optimizer. Tests establish exact base-weight equality after two adapter train/unload cycles. Held-out tests use either no examples or the full correct context. There is no checkpoint selection. This measures persistence on one language, not lifelong retention across unrelated tasks.

The primary Study B endpoint is the paired 0.5B clean-16 minus self-generated accuracy on compositions, across 24 environments, with a 95% t interval and 10,000-resample paired bootstrap sensitivity interval. All other intervals are descriptive. Comparing released model sizes confounds parameter count with their different pretraining histories; no causal scaling-law estimate is claimed. All model calls run locally on the same RTX 4090 D, with full query predictions, prompt hashes, token counts, measured times, and peak memory.

5.4 Study C: response-interface sensitivity

We freeze 24 additional environments (seeds 5000–5023) before formal inference and evaluate the same pinned 0.5B and 7B checkpoints. Each environment has the same 16 feedback, 16 direct, and 32 composition queries as Study B's design. The four conditions are no examples, 16 correct examples, 16 sequential guesses, and oracle offsets. No weights change in this study.

Each condition is evaluated with Study B's constrained first-token interface and with full generation. In the latter, both digit-only instructions are replaced by a request for brief reasoning followed by a final line `Answer: N`. Decoding is greedy with a 128-new-token cap and inherited checkpoint defaults: repetition penalties are 1.1 for 0.5B and 1.05 for 7B. After tokenizer special tokens are removed, the last nonempty line must match `Answer: [0-7]`, ignoring case and optional whitespace. Invalid formats count as wrong. Self-history stores the parsed digit or the literal string `invalid`. All output tokens through EOS, decoded text, invalid formats, truncations, input/output tokens, and measured time are retained.

The primary endpoint is the paired environment-level interaction for 0.5B composition accuracy,

$$\Delta_{\text{int}} = (A_{\text{clean}} - A_{\text{none}})_{\text{generation}} - (A_{\text{clean}} - A_{\text{none}})_{\text{first token}},$$

with a 95% t interval and a 10,000-resample bootstrap interval. Remaining intervals are descriptive. An interaction must be interpreted with its component means and within-interface feedback effects: it can be positive because the control deteriorates. The intervention changes instructions, output constraints, and compute jointly; it does not isolate the effect of reasoning tokens. Development uses seed 9 only. Prompt and cap remain fixed after its failures. A prose error that described both repetition penalties as 1.1 was corrected after launch; the frozen rule to inherit model defaults and the runner were unchanged. Exact source hashes and this amendment are retained.

6 Results

6.1 Adaptation and capacity

The small adaptive model exceeds the large frozen model by 52.91 points on linear tasks (97.5% CI [47.62,58.20]) and 37.32 on XOR ([28.85,45.79]). When both receive feedback, small-minus-large is 0.00 points for linear (95% CI [-0.22,0.22]) and -10.93 for XOR ([-15.39,-6.46]). The latter comparison shows a substantial remaining capacity effect under the tested update schedule. Logistic regression remains near chance on XOR despite clean feedback. Exact XOR is outside this fixed linear policy class, but observed chance performance also depends on the loss and optimization: a biased linear threshold can exceed chance under a zero-one objective. We do not identify chance as the best attainable linear-classifier accuracy or a limit for all small controllers.

Table 1: Mean post-change held-out accuracy (%), 20 seeds. All adaptive columns receive the same 8,192 clean examples; frozen models receive none.

Parameters	Linear frozen	Linear clean	XOR frozen	XOR clean
9	45.61	90.67	50.12	49.87
161	45.56	98.53	47.64	84.97
641	45.64	98.71	47.69	96.14
2,561	45.62	98.53	47.65	95.90

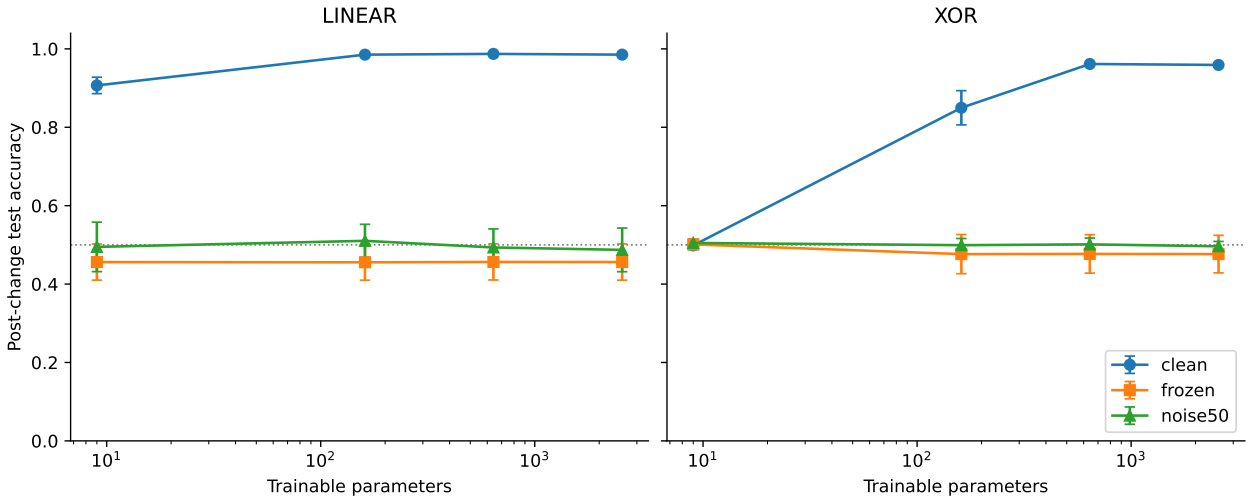


Figure 1: Capacity and feedback jointly determine observed accuracy. Error bars are descriptive 95% t intervals across 20 seeds. The dotted line is chance accuracy; noisy feedback can coexist with information retained from warmup, so individual noisy outcomes need not equal chance.

At an equal adaptation arithmetic proxy, the small model's 256 updates and the largest model's 16 updates both cost 7,077,888 proxy FLOPs. The small model leads by 42.80 points on linear tasks and 34.72 on XOR. However, it receives sixteen times as many new examples, while the large model's warmup cost is sixteen times greater. This is a particular compute/information trade, not an isolated causal effect of feedback-loop efficiency or a scaling law.

6.2 Feedback variations, transfer, and retention

For the 161-parameter model, clean, 25% corruption, and 50% corruption yield 98.53%, 92.05%, and 51.03% accuracy on linear tasks and 84.97%, 71.84%, and 49.92% on XOR. Inverted feedback yields 1.20% and 13.32% respectively. More updates can reinforce the wrong target when the update rule assumes trusted labels.

Delayed feedback reaches 98.28% and 81.87% at the online endpoint. After flushing, its scores and final training losses equal clean feedback for every seed and capacity. This equality is a correctness check for ordered updates, not an additional empirical discovery about latency.

Clean small-model uniform-input accuracy is 98.44% for linear rules and 88.19% for XOR. This is distribution transfer within the same known rule family, not transfer to an unrelated task. Old-rule accuracy after clean adaptation is 45.23% and 51.94%. Replay raises these to 54.92% and 64.43% while lowering new-rule accuracy to 89.85% and 76.48%. Because rules may conflict without a

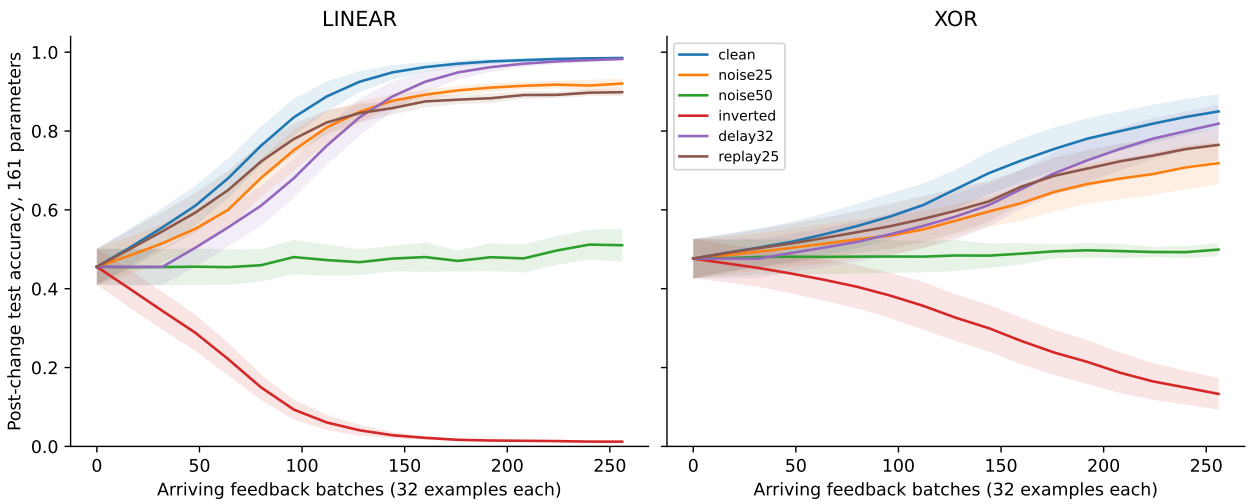


Figure 2: The same 161-parameter learner under six update conditions. Shading shows descriptive 95% seed-level intervals at fixed arrival checkpoints.

task identifier, this is an adaptation– retention tradeoff; perfect performance on both is not always possible.

6.3 Study B: feedback does not guarantee compositional gains

Table 2: Study B mean constrained first-token accuracy (%), 24 environments. Direct tests use one command; composition tests use two. Chance is 12.5%.

Model	Direct none	Direct clean	Compose none	Compose clean	Compose oracle
0.5B	13.80	12.24	11.46	11.46	11.46
1.5B	13.28	13.02	11.20	12.63	12.89
3B	10.16	5.73	13.02	12.63	14.32
7B	13.80	26.56	13.80	17.58	26.82

For the primary 0.5B comparison, clean feedback and self-generated examples both yield 11.46% composition accuracy. The paired difference is 0.00 points (95% t CI [-1.03,1.03]; bootstrap [-0.78,1.04]). Clean-minus-none is also 0.00 points. This is a failure to demonstrate the hypothesized gain with this mechanism, not proof that every feedback mechanism must fail.

The 7B model reaches 26.56% on direct tasks with clean-16 examples versus 13.80% without examples. Its composition scores are 17.58% and 13.80%, respectively: a descriptive paired difference of 3.78 points (95% CI [0.38,7.17]). With explicit offsets, direct accuracy rises to 75.52%, while composition reaches 26.82%. Representation of the instruction, inference of the latent rule, and execution of compositions are distinct bottlenecks.

LoRA trains 540,672 parameters in addition to the frozen 0.494B base. Without example context, its direct accuracy is 26.30% versus 13.80% for the base. The paired direct gain is 12.50 points (descriptive 95% t CI [3.76,21.24]). Composition accuracy is 13.15%; its paired improvement over no-feedback base is 1.69 points (95% CI [-2.12,5.50]). Retaining examples at inference yields 13.28% composition accuracy but reduces direct accuracy to 9.90%. Thus the persistent single-command

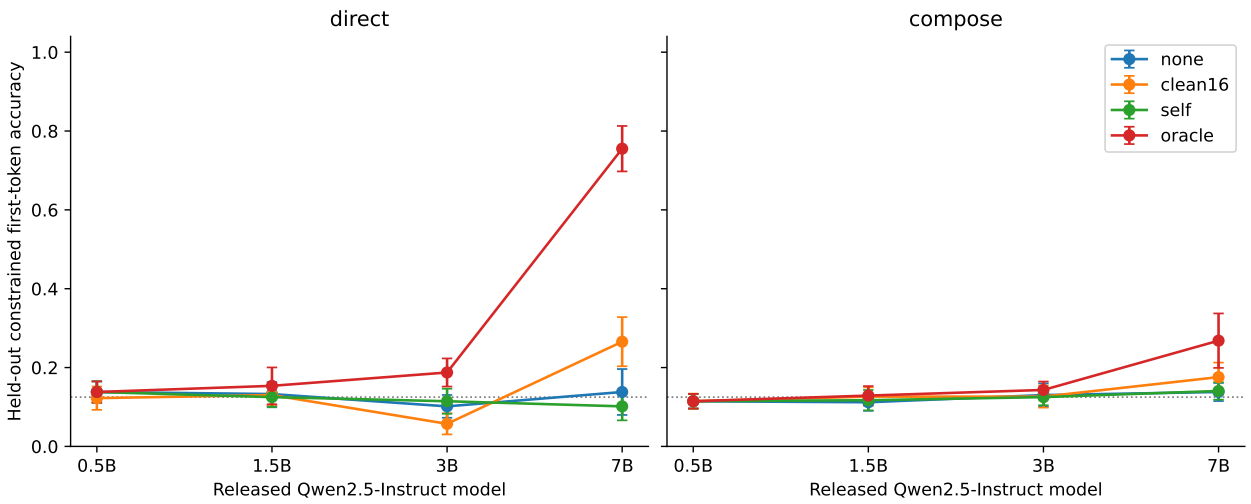


Figure 3: Released-model size and the supplied evidence affect direct and compositional performance differently. Error bars are descriptive environment- level 95% t intervals. The oracle provides extra structured information; it is not a cost-matched competitor.

gain is strongly context-dependent: adding the same demonstrations does not preserve it. These results distinguish a gain under the training prompt format from robust compositional transfer.

Each frozen-model suite processes 2,156,304 unpadded input tokens in 888 forward calls, including self-history construction and evaluation. Measured suite times are 21.81, 56.22, 109.02, and 234.69 seconds; peak allocated memory is 1.20, 3.41, 6.41, and 15.29 GiB. These figures are whole-suite accounting, not equal lifetime budgets or per-condition causal estimates. The LoRA suite adds 1,486,848 training input tokens and 86.67 measured update seconds. Its extra optimization is explicitly not cost-matched to frozen inference.

6.4 Study C: format and budget limit the generation comparison

All 9,216 held-out predictions and 768 self-history records per model pass source, model-revision, environment, completeness, raw-token decoding, and score checks. Table 3 reports composition accuracy; Figure 4 also shows direct predictions.

Model	Interface	No examples	Clean 16	Self	Oracle
0.5B	First token	10.68	11.20	10.68	10.68
0.5B	Generation	0.00	0.00	0.00	0.00
7B	First token	13.15	16.15	12.24	24.74
7B	Generation	0.00	6.90	0.26	97.27

Table 3: Study C composition accuracy (%). Generation scores include invalid and truncated responses as defined by the frozen parser; they must be read with the format and budget results below.

The primary 0.5B interaction is −0.52 percentage points (95% t interval [−1.16,0.11]; bootstrap [−1.17,0.00]). Its first-token clean-minus-none effect is 0.52 points, while the generation effect is exactly zero in this sample. Every generated answer is the literal `N` followed by EOS, including the oracle condition. This is a failure of the tested instructions and output interface; it cannot establish that the model lacks the reasoning capacity measured by a different interface.

For 7B, oracle generation reaches 100% direct and 97.27% composition accuracy, compared with 69.53% and 24.74% under first-token scoring. Thus the narrow first-token interface can substantially understate execution capability when the true rules are supplied. With examples, generation's clean-minus-none composition contrast is 6.90 points (descriptive 95% interval [4.04,9.76]), but absolute clean accuracy falls from 16.15% to 6.90%. Generation improves that contrast partly against a control with zero measured accuracy; it does not establish a superior overall inference method.

The 128-token cap is material. In 7B composition queries, truncation rates are 100%, 88.28%, 95.70%, and 0.26% for no examples, clean examples, self examples, and oracle offsets respectively. Corresponding invalid-format rates are 100%, 88.28%, 95.57%, and 0.26%. A truncated response can still contain a parseable final line, so these quantities need not coincide. Clean-example direct generation scores 24.22%, with 57.55% truncation. These outcomes leave longer-budget example induction unresolved; they are not evidence of an unrestricted inability to infer the offsets.

Self-history also deteriorates under the interface: all 384 generated examples for 0.5B and 354 of 384 for 7B are invalid. The resulting histories contain the literal `invalid`, as specified. They are not 16 valid critiques or error-directed repairs. The contrast measures the complete construction of this self-history under the fixed prompt and budget.

Whole-suite elapsed times, excluding loading, are 28.92 and 1,831.46 seconds. Each model uses 672 calls per interface, including history acquisition. The 7B first-token and generation paths take 133.83 and 1,695.16 measured forward seconds; generation emits 554,021 tokens, or 599,945 token slots with batch padding. Such costs are recorded, not matched. All generation defaults, including the model-specific repetition penalties, remain part of the treatment.

6.5 Limitations and interpretation

All three studies use synthetic tasks with disclosed families and inexpensive feedback. They do not measure autonomous exploration, long-horizon reward, human intelligence, or general cross-domain skill accumulation. Study A's fixed optimization schedule may interact with capacity. Its replay treatment also changes fresh-example exposure. Study B uses only one model family and a first-token response interface. Study C adds full generation but also a strict final-line format and a fixed token cap. Low accuracy can reflect the interface, instruction-following ability, prior execution ability, or the specific update schedule. None of these studies reproduces a full reflection and error-repair agent with comparable inference-cost controls.

In development, 0.5B answered a simple arithmetic question incorrectly under a one-digit instruction. The official weight SHA-256 matched, and CPU float32, GPU float32/bfloat16, and eager/SDPA paths reproduced it. A subsequent ordinary question produced the correct answer; other ordinary-language responses were coherent. Thus the calibration reveals prompt sensitivity, not general arithmetic incapacity. Model files and device agreement alone cannot certify that a response interface makes full use of the available capability.

The initial Study A random-stream defect and the corrected fresh-seed replication are both disclosed and preserved. A disk-full error occurred only during Study B export; every original run had completion metadata and complete predictions. Export was repeated on a separate filesystem without rerunning models. Source hashes and complete outcome grids were checked before analysis.

Prior work already establishes reinforcement learning, external memory, verbal reflection, and test-time computation as relevant mechanisms. Our current evidence is a diagnostic investigation of the stated hypothesis, not a first discovery of feedback-driven intelligence. Full-text review of five close precedents strengthens the comparison, but a systematic novelty review remains incomplete. External memory compression, active exploration, sustained cross-task retention, and long-horizon reward learning remain untested.

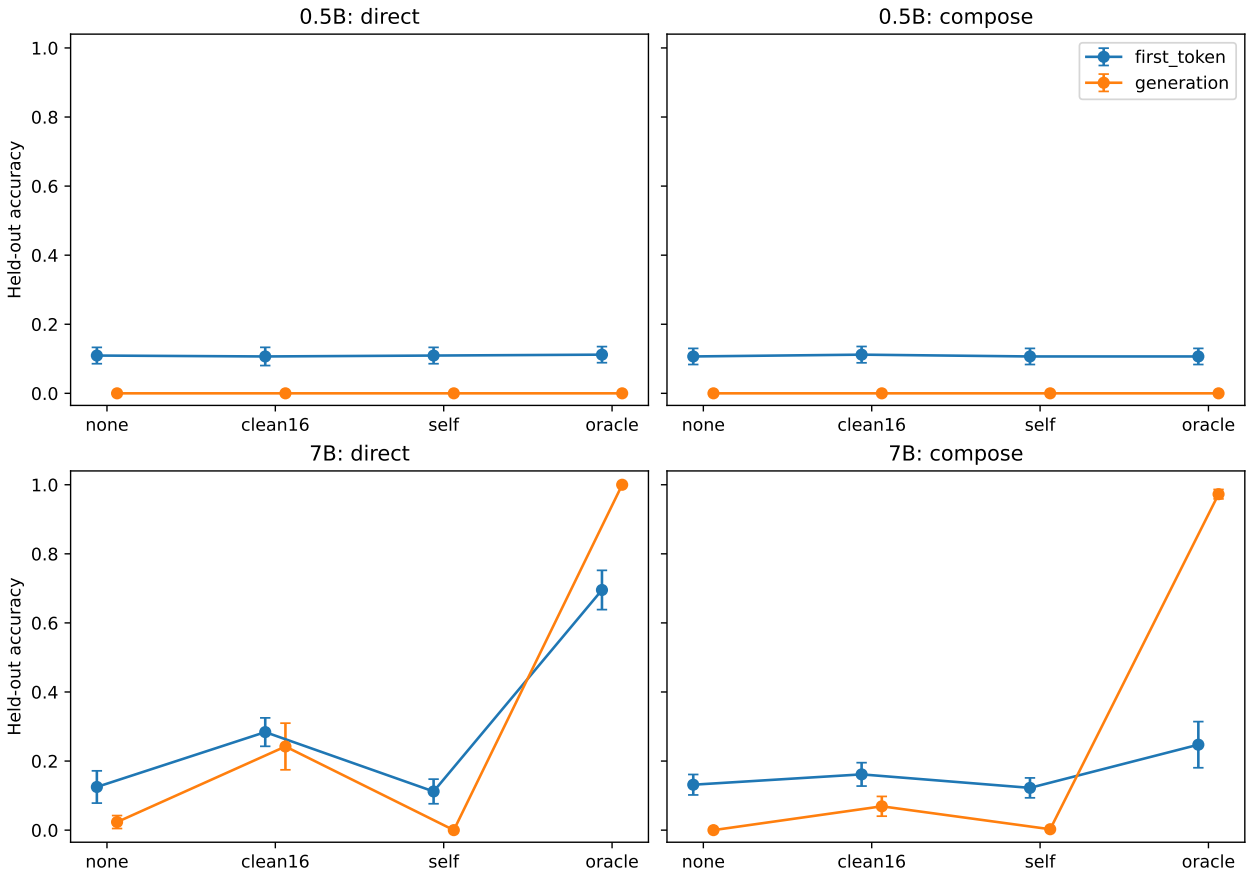


Figure 4: Study C response interfaces. Error bars are descriptive 95% intervals across 24 environment means. Strong oracle execution under generation coexists with pervasive format or token-cap failures in other conditions.

7 Conclusion

Feedback enables substantial adaptation in small neural learners under task change, but capacity remains consequential when both small and large models learn. Retaining correct examples does not yield the hypothesized compositional improvement in our smallest pretrained model under first-token scoring. A short parameter-update schedule yields a persistent direct gain without clear compositional transfer. Full generation reveals substantially greater oracle execution capability in the larger model, while format failures and truncation prevent a broad inference about the limits of example-based feedback.

The supported conclusion is conditional: feedback, representation, prior learning, update rules, and response interfaces jointly determine measured adaptation. Further evidence must test actual repeated error-directed updates, adequate generation budgets, retention, and transfer under comparable resource constraints. The present results do not establish that feedback alone explains intelligence or that increasing model capacity is an incorrect direction.

References

[1] D. Silver, S. Singh, D. Precup, and R. S. Sutton. Reward is enough. Artificial Intelligence, 299:103535, 2021. https://doi.org/10.1016/j.artint.2021.103535.

[2] A. Vaswani et al. Attention Is All You Need. NeurIPS, 2017. https://arxiv.org/abs/1706.03762.

[3] J. Kaplan et al. Scaling Laws for Neural Language Models. 2020. https://arxiv.org/abs/2001.08361.

[4] J. Hoffmann et al. Training Compute-Optimal Large Language Models. 2022. https://arxiv.org/abs/2203.15556.

[5] H. Chen et al. Feedback is all you need: from ChatGPT to autonomous driving. 2023. https://doi.org/10.1007/s11432-023-3740-x. Bibliographic metadata verified; full-text argument not assessed in this version.

[6] Y. Duan et al. RL^2: Fast Reinforcement Learning via Slow Reinforcement Learning. 2016. https://arxiv.org/abs/1611.02779v2.

[7] M. Laskin et al. In-context Reinforcement Learning with Algorithm Distillation. 2022. https://arxiv.org/abs/2210.14215.

[8] N. Shinn et al. Reflexion: Language Agents with Verbal Reinforcement Learning. 2023. https://arxiv.org/abs/2303.11366v4.

[9] A. Madaan et al. Self-Refine: Iterative Refinement with Self-Feedback. 2023. https://arxiv.org/abs/2303.17651v2.

[10] J. Huang et al. Large Language Models Cannot Self-Correct Reasoning Yet. ICLR, 2024. https://arxiv.org/abs/2310.01798v2.

[11] C. Snell et al. Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters. 2024. https://arxiv.org/abs/2408.03314.